

TWEEN BEHAVIOR SCRIPT INTERFACE

The `ITweenBehaviorInstance` interface derives from `IBehaviorInstance` to add APIs specific to the **Tween behavior**.

An actively running tween is represented by `ITweenState`, which also derives from `ITimelineState`. These interfaces can be used to control playback, including identifying when tweens end via the `finished` promise.

Examples

See the **Scripting tweens example** for a demonstration of using tweens from JavaScript code.

Basic tween usage

A code sample is shown below of starting a tween and waiting for it to finish.

```

1  async function doTween(runtime)
2  {
3      // Get a Sprite instance with the Tween behavior
4      const inst = runtime.objects.Sprite.getFirstInstance();
5
6      // Create a tween that moves it to (300, 300) over 2 seconds
7      const tween = inst.behaviors.Tween.startTween("position", [300, 300], 2);
8
9      // Wait for the tween to finish
10     await tween.finished;
11
12     // Log to the console now the tween has finished
13     console.log("Tween finished");
14 }
```

More examples

Some examples of valid calls to `startTween` are shown below (assuming `Tween` represents this behavior).

```
1 // Tween X position to 300 over 2 seconds linearly
2 Tween.startTween("x", 300, 2, "linear");
3
4 // Tween position to (300, 300) over 2 seconds with ease "in-out-sine"
5 Tween.startTween("position", [300, 300], 2, "in-out-sine");
6
7 // Looping ping-pong tween to size 200x200 every 0.5 seconds
8 Tween.startTween("size", [200, 200], 0.5, "out-sine", {
9   loop: true,
10  pingPong: true
11 });
12
13 // Tween color to blue over 1.5 seconds linearly
14 Tween.startTween("color", [0, 0, 1], 1.5, "linear");
15
16 // Value tween from 100 to 200 linearly over 3 seconds
17 const t = Tween.startTween("value", 200, 3, "linear", {
18   startValue: 100
19 });
20 // (then read t.value over time)
```

Tween properties

When using the `startTween` method, the `prop` parameter must be one of the strings given in the table below. Each property also lists how many values are expected for the `endValue` parameter; if more than 1, they should be passed as an array.

Property	Number of values
"x"	1
"y"	1
"position"	2
"z"	1
"width"	1
"height"	1
"x-scale"	1
"y-scale"	1
"size"	2
"scale"	2

Property	Number of values
"angle"	1 (in radians)
"opacity"	1 (in 0-1 range)
"color"	3 (RGB values in 0-1 range)
"value"	1

Ease names

When using the `startTween` method, the `ease` parameter must be one of the strings given in the table below, or the name of a custom ease in the project.

"linear"		
"in-sine"	"out-sine"	"in-out-sine"
"in-elastic"	"out-elastic"	"in-out-elastic"
"in-back"	"out-back"	"in-out-back"
"in-bounce"	"out-bounce"	"in-out-bounce"
"in-cubic"	"out-cubic"	"in-out-cubic"
"in-quadratic"	"out-quadratic"	"in-out-quadratic"
"in-quartic"	"out-quartic"	"in-out-quartic"
"in-quintic"	"out-quintic"	"in-out-quintic"
"in-circular"	"out-circular"	"in-out-circular"
"in-exponential"	"out-exponential"	"in-out-exponential"

Tween behavior APIs

`startTween(prop, endValue, time, ease, opts)`

Start a tween running for a property to a given end value, over a `time` given in seconds, with an ease function specified by `ease`. Returns an `ITweenState` representing the running tween.

- `prop` must be a string of one of the property names given in the table in the section *Tween properties* above.

- **endValue** must be either a number, or an array of numbers, depending on **prop**. In the table of properties above, where the *Number of values* is 1, this must be a number; where it is greater than 1, it must be an array with that many values.
- **time** is the duration the tween will run for in seconds.
- **ease** is a string of the name of one of the built-in eases in the section *Ease names* above, or the name of a custom ease in the project.

The **opts** parameter is optional for providing further parameters via object properties. The following properties can be used:

- **tags**: a list of tags to assign to the tween, specified either as a space-separated string, or an array of strings
- **destroyOnComplete**: a boolean indicating whether to automatically destroy the instance once the tween completes (default false)
- **loop**: a boolean indicating whether to repeat the tween when it reaches the end (default false)
- **repeatCount**: the number of times to repeat the tween (default 1).
- **pingPong**: a boolean indicating whether to alternate the playback direction when repeating (default false)
- **startValue**: for value tweens only, specifies the start value (default 0).

See above for some code examples demonstrating some of the ways this method can be called.

***allTweens()**

Iterates all actively running tweens created by the behavior, represented with [ITweenState](#).

***tweensByTags(tags)**

Iterates all actively running tweens matching the given set of tags, represented with [ITweenState](#). The tags may be specified as either a space-separated string, or an array of strings.

isEnabled

A boolean indicating if the behavior is enabled. If disabled, the behavior no longer has any effect on the object.

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